

NATIONAL JUNIOR COLLEGE
SH2 PRELIMINARY EXAMINATION
Higher 1

CANDIDATE
NAME

SUBJECT
CLASS

REGISTRATION
NUMBER

CHEMISTRY

Paper 1 Multiple Choice

Additional Materials:

Optical Answer Sheet
Data Booklet

8873/01

Thur 13 September 2018

1 hour

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, subject class and registration number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **30** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

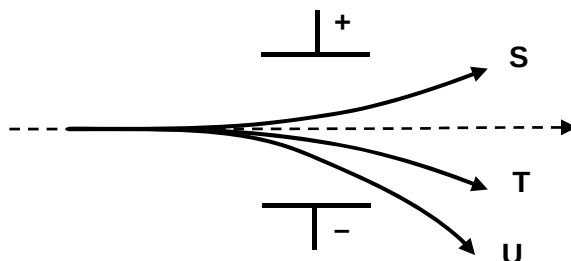
The use of an approved scientific calculator is expected, where appropriate.

This paper consists of **14** printed pages

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1** Use of the Data Booklet is relevant to this question.

The following are flight paths of charged particles when accelerated in an electric field.

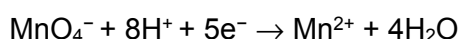


Which of the following correctly identifies **S**, **T** and **U**?

	S	T	U
A	$^{15}\text{O}^+$	$^{14}\text{C}^+$	$^{14}\text{N}^+$
B	$^{15}\text{O}^-$	$^{15}\text{O}^+$	$^{28}\text{Si}^+$
C	$^{14}\text{N}^-$	$^{28}\text{Si}^{2+}$	$^{14}\text{C}^{2+}$
D	$^{14}\text{N}^-$	$^{14}\text{C}^+$	$^{28}\text{Si}^{2+}$

- 2** An element **L** can exist in a few oxidation states. 0.1 mol of L^{2+} requires 0.04 mol of acidified KMnO_4 for complete reaction.

The half equation for reduction of MnO_4^- is



What is the final oxidation state of **L**?

- A** +4 **B** +3 **C** 0 **D** -1

- 3** Use of Data Booklet is relevant to this question.

Nickel makes up 20% of the total mass of a coin. The coin has a mass of 10.0 g.

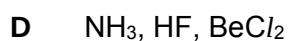
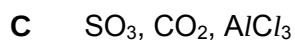
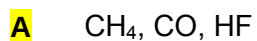
How many nickel atoms are in the coin?

- A** 2.05×10^{22} **B** 4.30×10^{22} **C** 1.03×10^{23} **D** 1.20×10^{24}

4 Which compound has an unpaired electron in its structure?



5 In which sequence is the molecules listed in the order of increasing dipole moment?

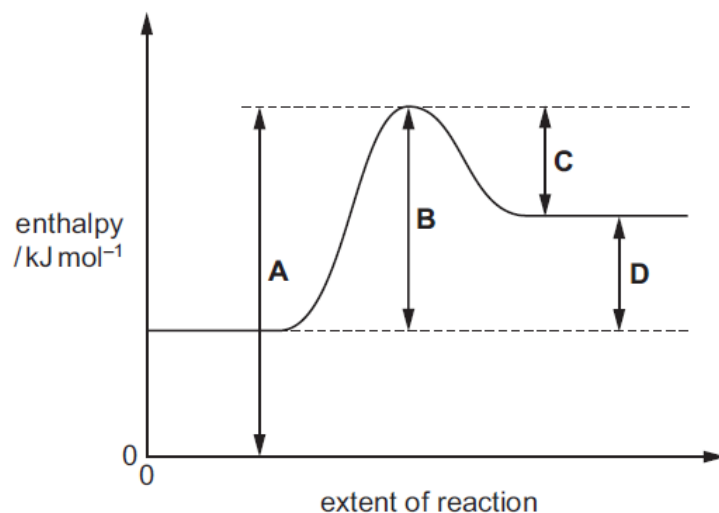


6 The C_2H_2 molecule is linear.

What can be deduced from this about the numbers of σ and π bonds present in the molecule?

	σ	π
A	2	2
B	2	3
C	3	1
D	3	2

7 The diagram shows a reaction pathway for an endothermic reaction.



Which arrow represents the activation energy for the forward reaction?

- 8 The standard enthalpy changes of combustion of carbon are as follows.

$$\text{C}(\text{graphite}) = -393 \text{ kJ mol}^{-1}$$

$$\text{C}(\text{diamond}) = -395 \text{ kJ mol}^{-1}$$

Which deductions can be made from the data above?

- 1 Graphite is more stable than diamond.
- 2 Graphite has a higher energy content than diamond.
- 3 Graphite is formed endothermically from diamond.

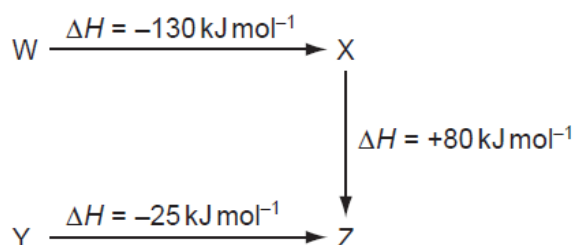
A 1, 2 and 3

B 2, and 3

C 1 and 3 only

D 1 only

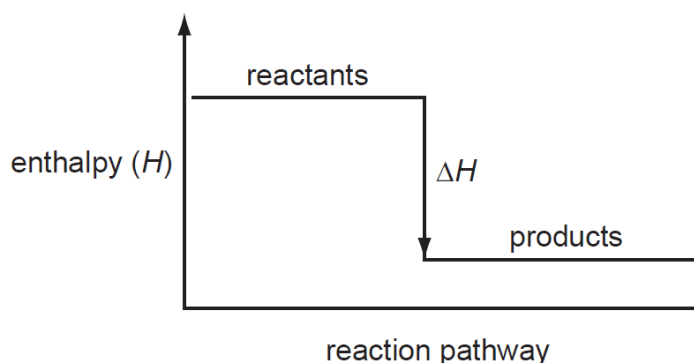
- 9 The diagram represents the energy changes for some reactions.



What are the natures of the conversions $W \rightarrow Y$, $Y \rightarrow X$ and $Z \rightarrow W$?

	$W \rightarrow Y$	$Y \rightarrow X$	$Z \rightarrow W$
A	exothermic	endothermic	endothermic
B	exothermic	exothermic	endothermic
C	endothermic	exothermic	exothermic
D	endothermic	endothermic	exothermic

- 10 Which enthalpy change could **not** be correctly represented by the enthalpy diagram shown?



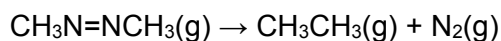
A standard enthalpy change of atomisation

B standard enthalpy change of combustion

C standard enthalpy change of hydration

D standard enthalpy change of neutralisation

- 11 Azomethane decomposes when heated according to the following equation.



This reaction is first order with respect to azomethane.

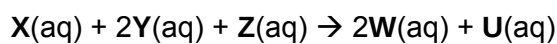
In experiment I, 160 cm³ of azomethane took 20 min to decompose to 10% of its original volume measured at constant temperature and pressure.

In experiment II, 320 cm³ of azomethane was allowed to decompose to 10% of its original volume at the same temperature and pressure as experiment I.

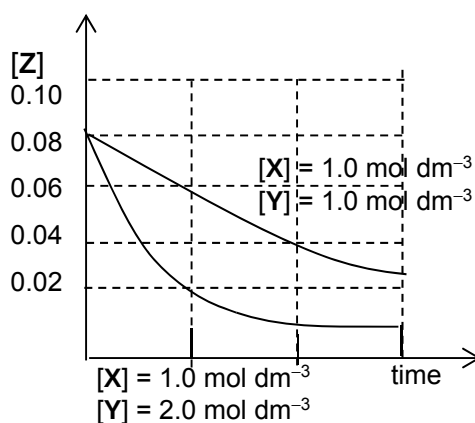
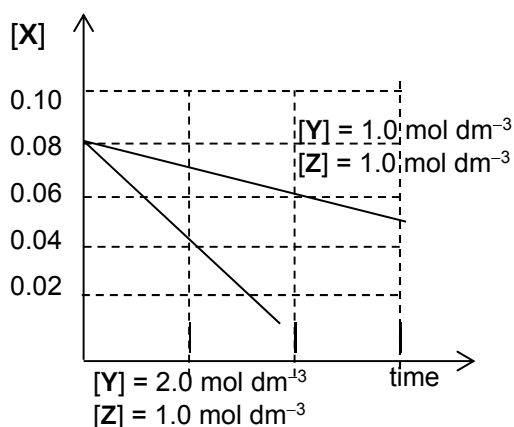
What is the time taken for 320 cm³ of azomethane to decompose to 10% of its original volume?

- A 10 min
- B 20 min**
- C 30 min
- D 40 min

- 12 Substances X, Y and Z react according to the following equation:



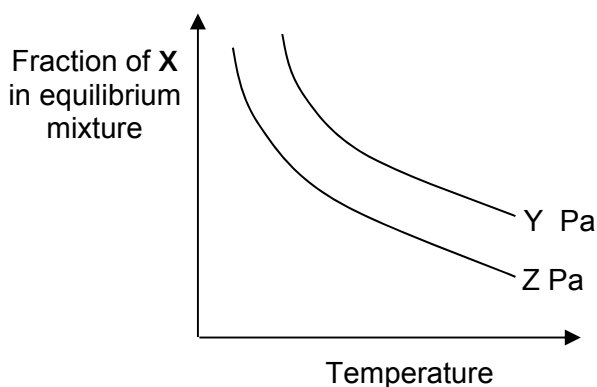
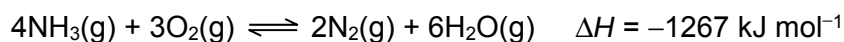
To find the rate equation for the above reaction, two sets of separate experiments were performed, in which the initial concentrations of each of the reactants X, Y and Z were varied while the other two were kept constant. The results are shown below.



Which rate equations represent the experimental results?

- A rate = $k[\text{Y}]^2$
- C rate = $k[\text{Y}]^2[\text{Z}]$**
- B rate = $k[\text{Y}][\text{Z}]$
- D rate = $k[\text{X}][\text{Y}][\text{Z}]$

- 13** The graph below shows how the fraction of **X**, which represents one of the following compounds in the equilibrium mixture shown below, varies with temperature at pressures of Y Pa and Z Pa.



Identify **X** and the correct relative magnitudes of Y and Z.

	X	Pressure
A	N₂	Z > Y
B	O ₂	Y > Z
C	H ₂ O	Y > Z
D	NH ₃	Z > Y

- 14** One of the key production stages in the Contact Process is the production of sulfur trioxide.

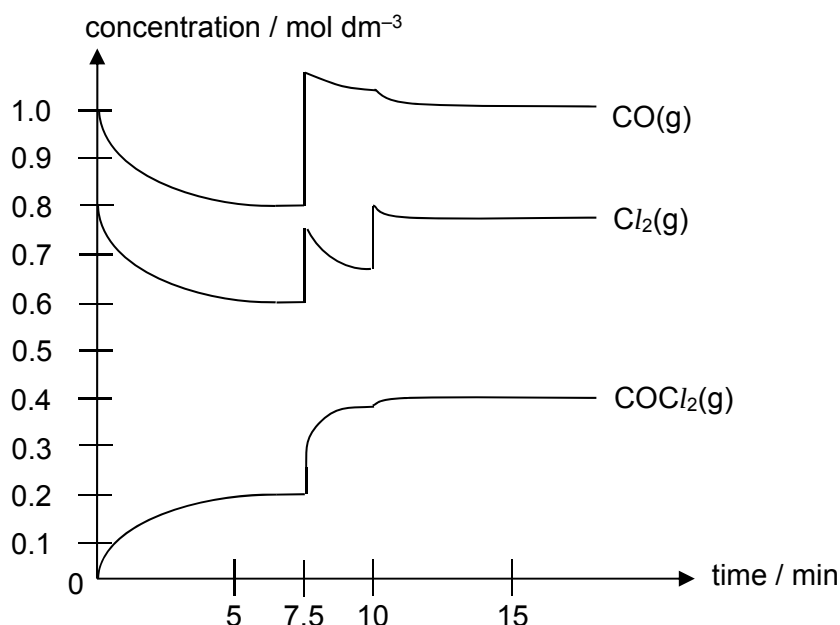
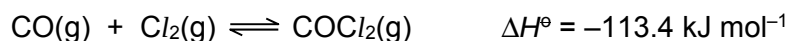


The rate constants of the forward and backward reactions are given as k_1 and k_{-1} respectively.

What happens to k_1 , k_{-1} and K_c if the temperature of the reaction is increased?

	k_1	k_{-1}	K_c
A	increase	increase	no change
B	increase	decrease	increase
C	decrease	increase	decrease
D	increase	increase	decrease

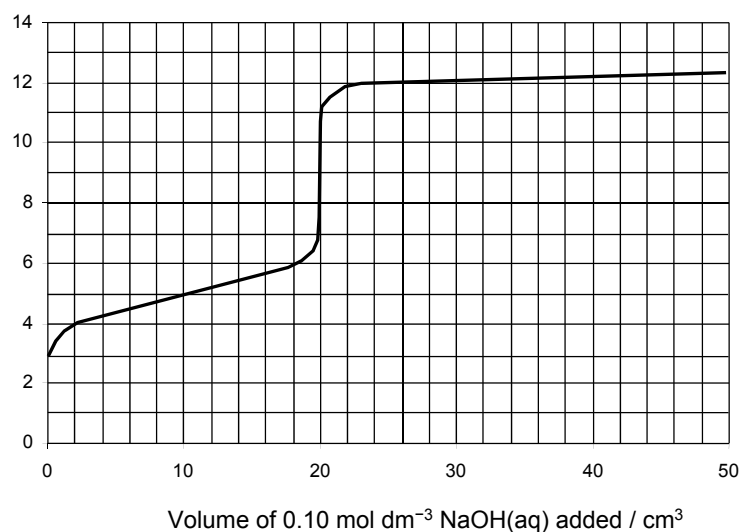
- 15** The reaction between carbon monoxide and chlorine was studied in an experiment by mixing the two gases and changing the reaction conditions inside the reaction vessel at different times during the experiment. The concentrations of the gases in the vessel were followed with time, and the following graph is obtained.



Which conclusion deduced from the graph is **incorrect**?

- A** The rate of forward reaction equals the rate of backward reaction at 7 min.
- B** The equilibrium constant, K_c , for the system when determined at 7 min is $2.4 \text{ mol}^{-1} \text{ dm}^3$.
- C** The change in concentration from 7.5 min to 10 min was produced by an decrease in volume at constant temperature.
- D** The change in concentration from 10 min to 15 min was produced by the addition of more chlorine.

- 16 The titration curve below shows the reaction between 20.0 cm³ of HX, and aqueous sodium hydroxide.



Given the following data:

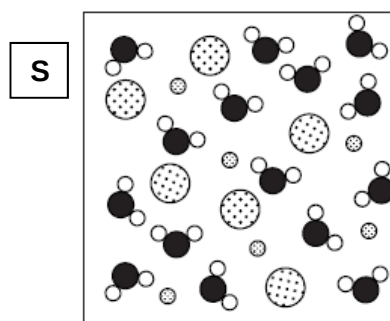
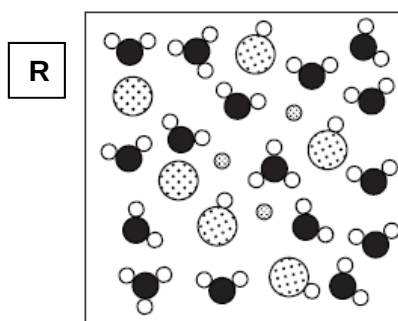
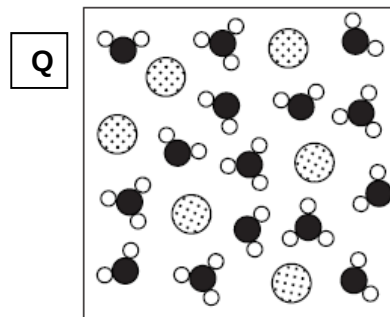
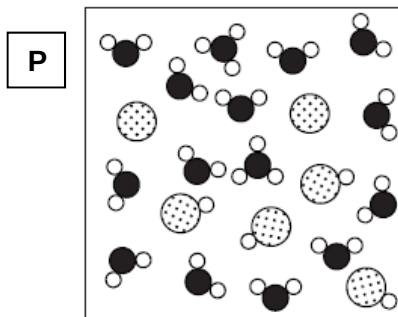
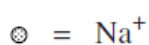
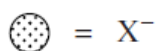
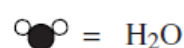
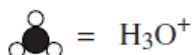
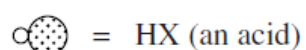
Indicator	Colour change (acidic to basic medium)	pH range in which colour change occurs
Methyl red	red to yellow	4.2 – 6.3
Bromothymol blue	yellow to blue	6.0 – 7.6
Thymolphthalein	colourless to blue	8.3 – 10.5

What statements are correct?

- 1 Methyl red is the most suitable indicator to be used for this titration.
- 2 Initial concentration of HX is 0.10 mol dm⁻³.
- 3 When 10.0 cm³ of NaOH is added, [H⁺] in the mixture is 1.00 × 10⁻⁵ mol dm⁻³.

A 1, 2 and 3 **B 2, and 3** C 1 and 3 only D 3 only

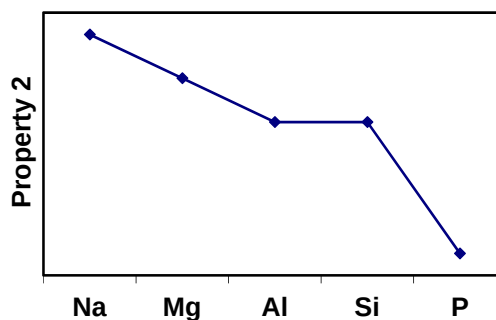
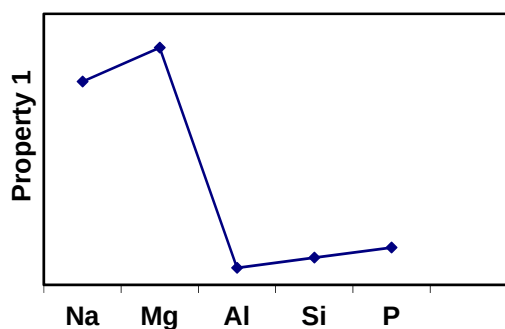
- 17 The diagrams represent some dilute aqueous solutions.
In all cases, only a few of the large numbers of water molecules are shown.



Which option correctly describes the solutions represented by the diagrams?

	P	Q	R	S
A	a strong acid	a weak acid	a salt formed by neutralising a strong acid with sodium hydroxide	weak acid partially neutralised by sodium hydroxide
B	a weak acid	a strong acid	a salt formed by neutralising a strong acid with sodium hydroxide	weak acid partially neutralised by sodium hydroxide
C	a strong acid	a weak acid	weak acid partially neutralised by sodium hydroxide	a salt formed by neutralising a strong acid with sodium hydroxide
D	a weak acid	a strong acid	weak acid partially neutralised by sodium hydroxide	a salt formed by neutralising a strong acid with sodium hydroxide

- 18 The graphs below show the variation of two properties of some period 3 elements and/or their compounds.



Which correctly describes properties 1 and 2?

Property 1

Property 2

- | | | |
|----------|--|---|
| A | atomic radius of the elements | electrical conductivity of the elements |
| B | boiling point of the highest oxidation state chlorides | pH of the highest oxidation state oxides when added to water |
| C | melting point of the highest oxidation state oxides | first ionisation energies of the elements |
| D | electrical conductivity of elements | pH of the highest oxidation state chlorides when added to water |

- 19 Element **H** is in Period 3 of the Periodic Table. The following four statements were made about the properties of element **J** or its compounds.

Three statements are correct descriptions and one is false.

Which statement does **not** fit with the other three?

- A** Adding NaOH(aq) to the solution resulting from the reaction of a chloride of **J** with water produces a white precipitate which is soluble in an excess of NaOH(aq).
- B** Element **H** is a solid at room temperature.
- C** Element **H** has the highest melting point among the period 3 elements.
- D** The oxide of element **H** is soluble in hydrochloric acid.

- 20 J, K and L are elements in Period 3 of the Periodic Table.

A mixture containing the oxides of J, K and L was dissolved in excess dilute sulfuric acid and filtered. The oxide of L was collected as a residue. When excess dilute sodium hydroxide was added to the filtrate, only a white precipitate of the hydroxide of K was formed.

What are the possible identities of J, K and L?

	J	K	L
A	Mg	Al	P
B	Al	Mg	P
C	Mg	Al	Si
D	Al	Mg	Si

- 21 Which are the principal factors which cause the properties of nanomaterials to differ significantly from other materials?

- 1 Size distribution
- 2 Specific surface feature
- 3 Very high surface area to volume ratio

- A 1, 2 and 3 B 1 and 2
C 2 and 3 D 1 only

- 22 The size of nanoparticles is between _____ m.

- A 10^{-8} to 10^{-7} B 10^{-10} to 10^{-8} C 10^{-9} to 10^{-7} D 10^{-11} to 10^{-9}

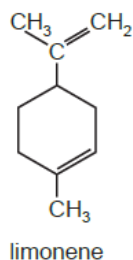
- 23 How many structural isomers can C_4H_9Cl form?

- A 3 B 4 C 5 D 6

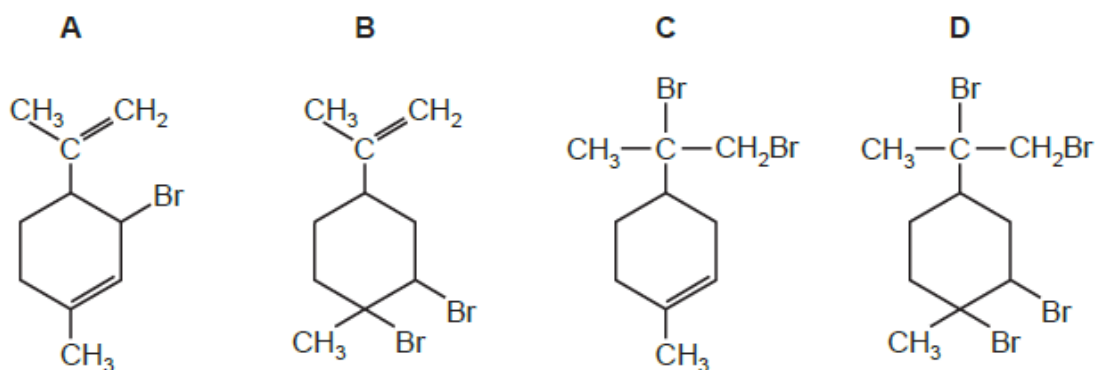
- 24 Compound G has the following structure: $(CH_3)_2C=CHCH_2CH(OH)CH_2CH=CH(CH_3)$
How many cis-trans isomers does Compound G have?

- A 0 B 2 C 3 D 4

- 25 Limonene is an oil formed in the peel of citrus fruits.



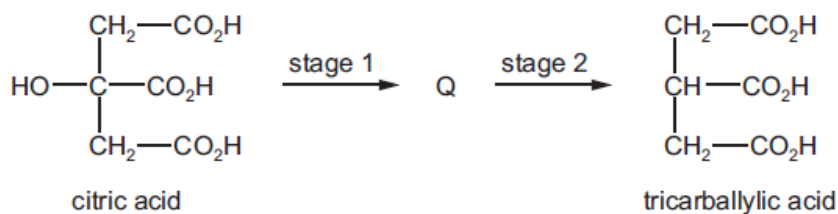
Which product is formed when an excess of bromine, $\text{Br}_2(\text{l})$ reacts with limonene at room temperature in the dark?



- 26 Which compound is a product of the hydrolysis of $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}_3$ by boiling aqueous sodium hydroxide?

A CH_3OH **B** $\text{C}_3\text{H}_7\text{OH}$ **C** $\text{C}_3\text{H}_7\text{CO}_2\text{H}$ **D** $\text{C}_3\text{H}_7\text{CO}_2^-\text{Na}^+$

- 27 Citric acid can be converted into tricarballic acid in two stages. An intermediate, Q, is formed.



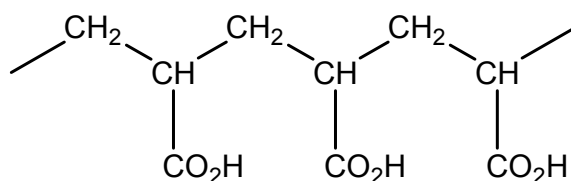
Which reagents are needed for each stage?

stage 1	stage 2
A Concentrated H_2SO_4	$\text{H}_2(\text{g})$ and Ni
B Concentrated H_2SO_4	LiAlH_4
C LiAlH_4	$\text{H}_2\text{SO}_4(\text{aq})$
D $\text{NaOH}(\text{aq})$	$\text{H}_2(\text{g})$ and Ni

28 Which statements are correct for monomers and polymers involved in addition polymerization?

- 1 The monomers contain at least one unsaturated bond.
 - 2 Empirical formula of the addition polymer is the same as that of its monomer.
 - 3 Poly(vinyl alcohol) is formed via addition polymerization and can form hydrogen bonds between its adjacent molecules.
 - 4 Low density poly(ethene) (LDPE) and high density poly(ethene) (HDPE) are addition polymers but differ in their monomers.
- A All correct.
- B 1, 2 and 3
- C 1 and 2
- D 1, 2 and 4

29 The absorbent material in babies' disposable nappies is made from the polymer



From which monomer could this polymer be obtained?

- A $\text{C/CH}_2\text{CH}_2\text{CO}_2\text{H}$
- B $\text{HOCH}_2\text{CH}_2\text{CO}_2\text{H}$
- C $\text{H}_2\text{C=CHCO}_2\text{H}$
- D $\text{HO}_2\text{CCH=CHCO}_2\text{H}$

30 Chlorofluorocarbons (CFCs) are commonly used as aerosols, propellants and refrigerants. However in the stratosphere, CFCs can damage the ozone layer through a radical chain reaction.

In which sequence are the following compounds listed in increasing order of their ability to destroy ozone?

- A $\text{CHClF}_2 < \text{CCl}_2\text{FCCl}_2 < \text{CHClF}_2$
- B $\text{CCl}_2\text{FCCl}_2 < \text{CHClF}_2 < \text{CHClF}_2$
- C $\text{CHClF}_2 < \text{CCl}_2\text{FCCl}_2 < \text{CHClF}_2$
- D $\text{CHClF}_2 < \text{CHClF}_2 < \text{CCl}_2\text{FCCl}_2$

End of Paper